AI Career Advisor – Prompt Engineering Project Report

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1. Objective

The purpose of this project is to build a basic **Career Advisor chatbot** using only **prompt engineering techniques**, without any external datasets, APIs, or fine-tuning. The chatbot should guide students by providing career paths, required skills, learning roadmaps, and resume tips.

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2. Prompt Design Approach

To make the model behave like a career advisor, three structured prompt components were used:

A. Role-Based Prompt

Defines the identity and behaviour of the AI.
It ensures the model consistently responds as a **professional career advisor**.

B. Instruction-Based Prompt

Provides rules and expectations for the model, such as:

- Give short and actionable advice
- Use bullet points for clarity
- Stay relevant to the user's background
- Avoid hallucination or incorrect information

C. Few-Shot Examples

Includes sample questions and high-quality answers to show the AI the correct response style.
This helps the model produce consistent, structured, and meaningful guidance.

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3. Before vs After Prompt Optimization

Before Tuning:

- Responses were unstructured and repetitive
- Some answers were generic or partially irrelevant
- Model confused topics and produced inconsistent suggestions

After Tuning:

- Answers became clear, structured, and technically accurate
- Guidance matched the user's query and academic background
- Responses followed bullet-point formatting
- No irrelevant or hallucinated content
- Model became more helpful and beginner-friendly

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4. Testing and Evaluation

The chatbot was tested with **multiple queries**, including:

- "I am from CSE. What career options are best?"
- "How should I start AI/ML as a beginner?"
- "What skills are needed to become a Data Analyst?"
- "How do I improve my resume?"

Outcome:
All responses were relevant, structured, and included proper skill paths, tools, and career steps. The chatbot delivered actionable guidance suitable for engineering students.

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5. Conclusion

This project shows that **prompt engineering alone** can create a functional and reliable career guidance chatbot. By combining role prompts, instructions, and example responses, the language model gave meaningful and accurate suggestions without requiring training data or complex systems. This approach is simple, cost-effective, and ideal for students learning the basics of Generative AI.

End of Report